

Supplement for RDD results

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Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- need to fix that plot
- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
- – robust rd not sig, donut RDD sig for 2 “larger holed donuts”, ordinary OLS is sig with 0.08 threshold,

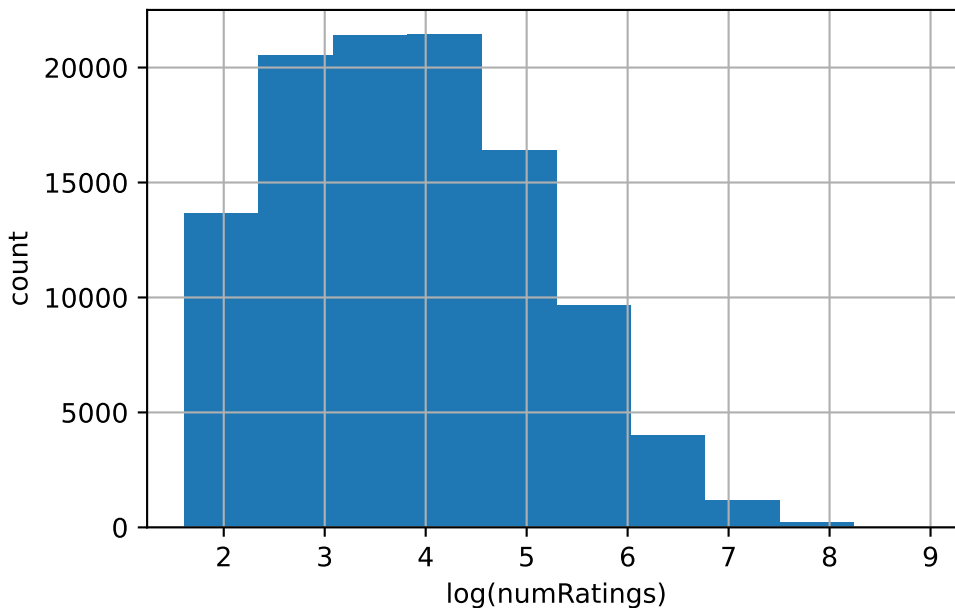
2 Setup & Data

```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 203392 notes written by the 3 major algorithms
Now we have 147890 notes classified as MISLEADING
Now we have 112758 notes which have enough ratings
Now we have 108602 aligned notes
--
FINAL total 108602 Notes
of which author writes again = 74108

NB Aligned data, current score
```

3 Plots

Most notes have a small number of rating, a small number have very many ratings This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting $\log(\text{numRatings})$ vs NoteIntercept, with colours according to Note final status (finalRatingStatus)



4 Trad OLS RDD

Just using OLS and an arbitrary window

bandwidth = 0.08

Number of treated units: 6897

Number of control units: 8167

Estimated Treatment Effect at the Cutoff: 0.0472

I think we can interpret this as a % change the chance of authoring again if published

P-value for treatment: 0.0057

T-statistic for treatment: 2.7647

5 RDD using rdrobust

This has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell,

Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

running variable (x) length 108602

binary outcome (y) length 108602

cutoff (c) is 0.4

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```
Number of Observations:      108602
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                      Triangular
Bandwidth Selection:           mserrd
Var-Cov Estimator:            NN
```

	Left	Right
Number of Observations	95363	13239
Number of Unique Obs.	95214	13223
Number of Effective Obs.	4382	4654
Bandwidth Estimation	0.054	0.054
Bandwidth Bias	0.094	0.094
rho (h/b)	0.577	0.577

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.014	0.024	0.587	5.573e-01	[-0.033, 0.061]
Robust	-	-	0.167	8.674e-01	[-0.05, 0.059]

MSE-optimal h = 0.0545

Bias-corrected point estimate: 0.0046 95% CI [-0.0496, 0.0588]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0272 (0.5x)	tau = -0.0180	95% CI [-0.1067, +0.0521]	N_L/N_R = 1687/2167
h = 0.0545 (1.0x)	tau = +0.0140	95% CI [-0.0496, +0.0588]	N_L/N_R = 4382/4654
h = 0.1089 (2.0x)	tau = +0.0389	95% CI [-0.0019, +0.0709]	N_L/N_R = 13640/9277

Mephi suggested text :

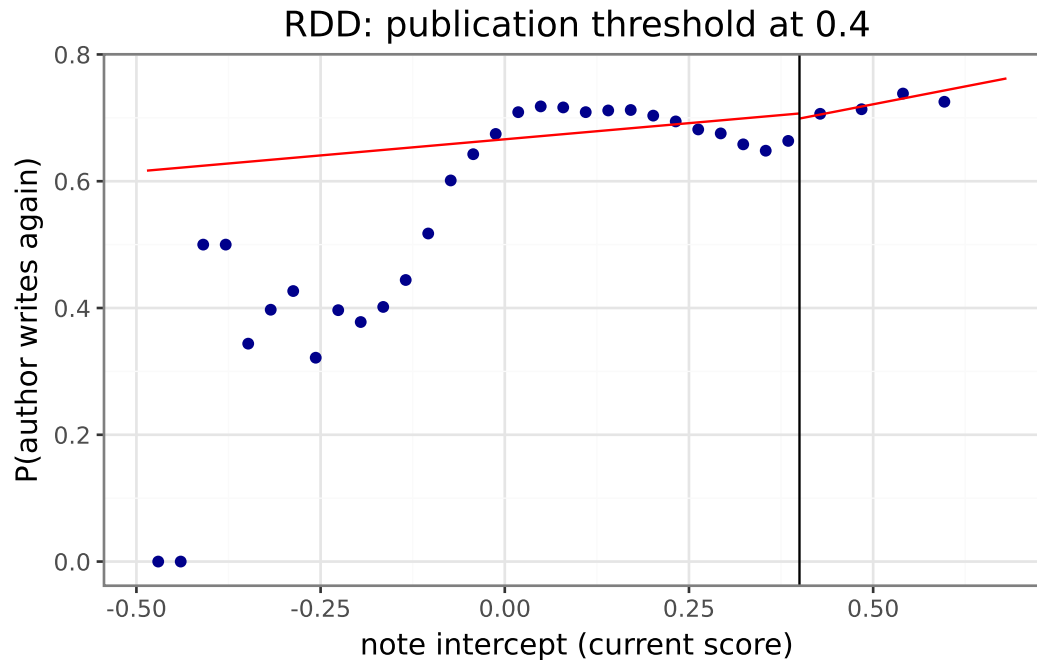
At the MSE-optimal bandwidth (h = 0.054) we estimate $\hat{\tau} = +1.4pp$ in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI [-5.0, +5.9], p = 0.87).

6 Converting rate to baseline outcome rate

Mephi suggested text: Untreated authors at the cutoff re-author at 65.7%; treated authors at the cutoff re-author at ~67.1%. Treatment effect = 1.4pp, a 2.1% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with gloabl linear fit (this is the default plot from `rdplot`)



```
['rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdplot_data']
paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (33 bins)
```

RD plot at $c = 0.4$ ($p=1$, triangular, $N=108602$)

bins used : 29 left / 5 right

J_IMSE : [29.0, 5.0]

J_MV : [868.0, 698.0]

scale : [1, 1]

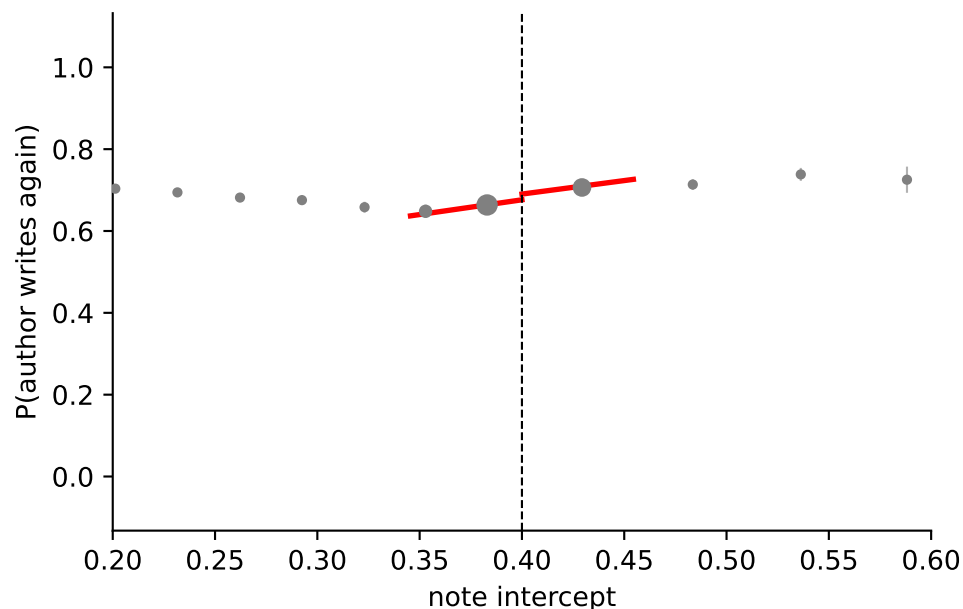
bin width : avg [0.03054120705045503, 0.056188602447509756], median [0.030541207050455044,

coefficients:

	Left	Right
0	0.706873	0.698851
1	0.101798	0.225296

wrote `paper-outputs/RDDrobust_scatter_meta.json`, `paper-outputs/RDDrobust_scatter_coef.csv`

Now with the local-linear fits (over the `rdrobust` defined window)



consistency check: gap at cutoff should conventional tau
 plot gap at $c = +0.0140$ vs conventional tau = $+0.0140$

8 Robustness check - density test

Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

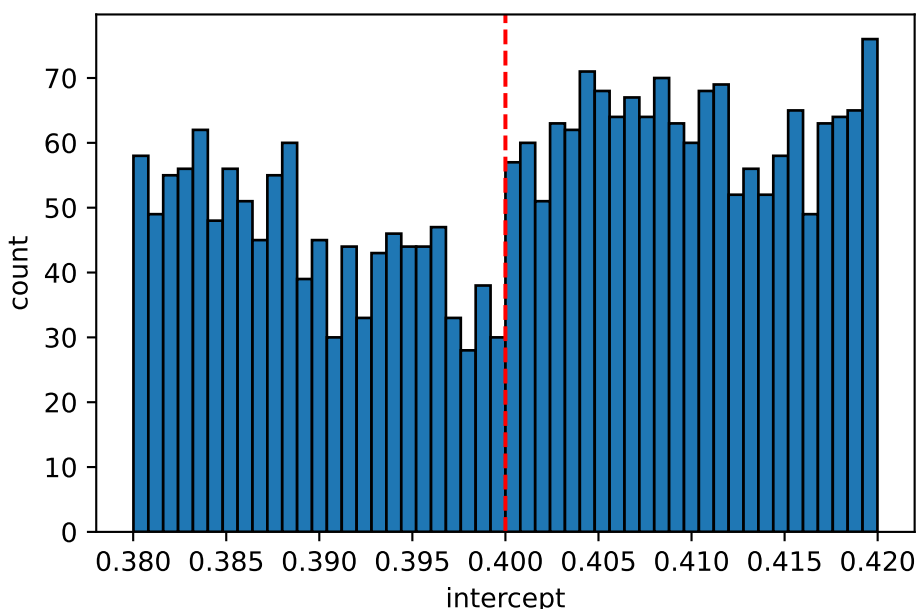
Inspecting density with a window of ± 0.02

```
t_asy      NaN
t_jk      5.511852e+00
p_asy      NaN
p_jk      3.550767e-08
dtype: float64
left      0.408601
right     0.756414
diff      0.347813
dtype: float64
left      0.048718
right     0.040207
dtype: float64
full      108602
left      95363
right     13239
eff_left   3745
eff_right  3336
dtype: int64
leftN      [1139.0, 1349.0, 1605.0, 1877.0, 2133.0, 2415....]
```

```

rightN      [1557.0, 1752.0, 1939.0, 2120.0, 2332.0, 2532....
leftWindow   [0.02, 0.023190887735418723, 0.026381775470837...
rightWindow  [0.02, 0.02224522414513161, 0.0244904482902632...
pval         [8.483401460315665e-16, 4.867596275664254e-13,...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino']

```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Excluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	4382	4654	0.054	0.0140	0.0239	0.8674	-0.0496	0.0588
± 0.005	6409	5674	0.071	0.0455	0.0237	0.1584	-0.0160	0.0981
± 0.010	8909	6737	0.087	0.0634	0.0223	0.0189	0.0107	0.1193
± 0.020	13338	8027	0.113	0.0571	0.0215	0.0498	0.0000	0.1142

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

```

running variable (x) length 108602
binary outcome (y) length 108602
cutoff (c) is 0.3

```

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 108602
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	83588	25014
Number of Unique Obs.	83460	24977
Number of Effective Obs.	13693	8767
Bandwidth Estimation	0.058	0.058
Bandwidth Bias	0.098	0.098
rho (h/b)	0.595	0.595

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.011	0.014	0.81	4.180e-01	[-0.016, 0.038]
Robust	-	-	0.916	3.595e-01	[-0.017, 0.046]

MSE-optimal h = 0.0581

Bias-corrected point estimate: 0.0145 95% CI [-0.0165, 0.0456]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0291 (0.5x) tau = +0.0244 95% CI [-0.0135, +0.0721] N_L/N_R = 6428/5071

h = 0.0581 (1.0x) tau = +0.0110 95% CI [-0.0165, +0.0456] N_L/N_R = 13693/8767

h = 0.1163 (2.0x) tau = +0.0020 95% CI [-0.0127, +0.0321] N_L/N_R = 30352/13033

running variable (x) length 108602

binary outcome (y) length 108602

cutoff (c) is 0.35

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 108602
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

Left	Right
------	-------

```

-----
Number of Observations      91478      17124
Number of Unique Obs.      91333      17104
Number of Effective Obs.   19489      8104
Bandwidth Estimation        0.1        0.1
Bandwidth Bias              0.15       0.15
rho (h/b)                   0.668     0.668

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional -0.006     0.014    -0.415    6.783e-01  [-0.033, 0.021]
Robust       -         -       -0.492    6.227e-01  [-0.04, 0.024]

```

MSE-optimal h = 0.1000

Bias-corrected point estimate: -0.0081 95% CI [-0.0403, 0.0241]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

```

h = 0.0500 (0.5x) tau = +0.0086 95% CI [-0.0304, +0.0627] N_L/N_R = 7889/3884
h = 0.1000 (1.0x) tau = -0.0057 95% CI [-0.0403, +0.0241] N_L/N_R = 19489/8104
h = 0.2000 (2.0x) tau = +0.0022 95% CI [-0.0245, +0.0235] N_L/N_R = 48080/15688

```

running variable (x) length 13239

binary outcome (y) length 13239

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```

Number of Observations:      13239
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                      Triangular
Bandwidth Selection:          mserrd
Var-Cov Estimator:           NN

```

```

-----
Left      Right
-----
Number of Observations      4223      9016
Number of Unique Obs.      4218      9005
Number of Effective Obs.    1181      1220
Bandwidth Estimation        0.013     0.013
Bandwidth Bias              0.022     0.022
rho (h/b)                   0.606     0.606

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional 0.08      0.041     1.967    4.919e-02  [0.0, 0.159]
Robust       -         -         1.96     4.996e-02  [0.0, 0.185]

```

MSE-optimal $h = 0.0131$

Bias-corrected point estimate: 0.0927 95% CI [0.0000, 0.1855]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0066 (0.5x) tau = +0.1160 95% CI [-0.0012, +0.2595] N_L/N_R = 605/659
h = 0.0131 (1.0x) tau = +0.0798 95% CI [+0.0000, +0.1855] N_L/N_R = 1181/1220
h = 0.0263 (2.0x) tau = +0.0507 95% CI [-0.0078, +0.1234] N_L/N_R = 2346/2358

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 13239
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	4223	9016
Number of Unique Obs.	4218	9005
Number of Effective Obs.	1181	1220
Bandwidth Estimation	0.013	0.013
Bandwidth Bias	0.022	0.022
rho (h/b)	0.606	0.606

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.08	0.041	1.967	4.919e-02	[0.0, 0.159]
Robust	-	-	1.96	4.996e-02	[0.0, 0.185]

All these are non-significant

11 Sensitivity check - data for different time windows

PERIOD 1, from 2025-04-08 to 2026-06-01

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 87791 first time note authors 2025-04-08 to 2026-06-01

Now we have 68536 notes written by the 3 major algorithms

Now we have 52423 notes classified as MISLEADING

Now we have 34862 notes which have enough ratings

Now we have 34000 aligned notes

--

FINAL total 34000 Notes
of which author writes again = 19008

NB Aligned data, current score
running variable (x) length 34000
binary outcome (y) length 34000
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 34000
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	29157	4843
Number of Unique Obs.	29147	4840
Number of Effective Obs.	2459	2383
Bandwidth Estimation	0.068	0.068
Bandwidth Bias	0.105	0.105
rho (h/b)	0.651	0.651

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.033	0.034	0.949	3.426e-01	[-0.035, 0.1]
Robust	-	-	0.648	5.173e-01	[-0.055, 0.109]

MSE-optimal h = 0.0683

Bias-corrected point estimate: 0.0271 95% CI [-0.0550, 0.1093]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0342 (0.5x) tau = +0.0286 95% CI [-0.0960, +0.1495] N_L/N_R = 884/1155
h = 0.0683 (1.0x) tau = +0.0327 95% CI [-0.0550, +0.1093] N_L/N_R = 2459/2383
h = 0.1366 (2.0x) tau = +0.0537 95% CI [-0.0110, +0.1006] N_L/N_R = 7268/4244

PERIOD 1, from 2024-01-01 to 2025-04-07

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 172844 first time note authors 2024-01-01 to 2025-04-07

Now we have 134572 notes written by the 3 major algorithms

Now we have 95257 notes classified as MISLEADING

Now we have 77741 notes which have enough ratings

Now we have 74450 aligned notes

--

FINAL total 74450 Notes
of which author writes again = 55026

NB Aligned data, current score
running variable (x) length 74450
binary outcome (y) length 74450
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 74450
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	66069	8381
Number of Unique Obs.	65977	8379
Number of Effective Obs.	3842	3485
Bandwidth Estimation	0.068	0.068
Bandwidth Bias	0.112	0.112
rho (h/b)	0.611	0.611

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.025	0.025	0.978	3.283e-01	[-0.025, 0.074]
Robust	-	-	0.519	6.035e-01	[-0.043, 0.074]

MSE-optimal h = 0.0682

Bias-corrected point estimate: 0.0154 95% CI [-0.0427, 0.0735]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0341 (0.5x)	tau = -0.0045	95% CI [-0.1048, +0.0660]	N_L/N_R = 1364/1625
h = 0.0682 (1.0x)	tau = +0.0246	95% CI [-0.0427, +0.0735]	N_L/N_R = 3842/3485
h = 0.1365 (2.0x)	tau = +0.0483	95% CI [+0.0060, +0.0846]	N_L/N_R = 12689/6865